

TANDY TRS-80 MODEL 2000 PERSONAL COMPUTER INTRODUCTION

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Tandy/Radio Shack has produced many innovative products containing the latest electronic technology, but the newest - the Tandy TRS-80 Model 2000 Personal Computer, has combined significant technology developments and state-of-the-art performance specifications to be Tandy's most outstanding technical achievement. To meet the developing market requirements of integrated productivity software with high resolution interactive graphics - monochrome and color, it was clear that we needed a high performance graphics oriented workstation. As we started defining this product technically, an unusual exciting thing happened - all the required technology improvements in semiconductors, CRT displays, disk drives, and so forth all fell into place at the right time and at the right price.

First we need a really fast microprocessor that has enough horsepower to perform demanding processes such as array calculations, record sorting, and document reformatting very quickly. I heard a PC user comment the other day - "I wish the recalculation of my cash flow spreadsheet would run out before my cash does." The new window environments using a bit map display like MS-Windows along with integrated productivity applications with graphics require a very fast microprocessor to have responsive screen interaction and acceptable through-put.

The clear choice is the latest 1983 Model - the INTEL 80186 Microprocessor. This device has very significant speed enhancements over the industry predominant 8088. The 16-bit data bus doubles the memory data rate. A dedicated hardware adder for effective address calculations results in three to six times faster execution. Specific instructions such as multiply, divide, and string move are 2 to 3 times faster even at the same clock speed. The 80186 operates at 8 MHz with no wait states in the Tandy 2000. That means the CPU does not have to wait additional clock cycles for the memory to be fetched.

Not only is it much faster, the 80186 provides a lower total cost by integrating 15 to 20 support chips into the one chip. These new functions include: Clock Generator, Multilevel Interrupt Controller, three Programmable Timers, and two High speed DMA Controllers. The speed of the 80186 is supported by a fully interrupt driven architecture, 150 nanosec 64K x 1 RAM devices, and by "F Parts," a new high speed family of logic chips. An extra parity check bit for memory error detection is implemented for all memory configurations from 128K to 768K bytes.

It is clearly evident by the exhibits at this conference that high resolution graphics have become a necessity in workstations, not only graphic presentations of the results, but moreover, graphics are key to the new applications environments to improve man-machine communication. The predominant resolution in this type product today is 640 x 200 because of cost reasons. Three times less resolution in one direction is obviously not desirable. We have doubled the number of horizontal pixels to 400 while maintaining 60 Hz refresh without interlace to avoid interlace flicker. Interlace flicker produces a distracting throbbing image when certain bit patterns are displayed. To accomplish this high resolution, the video RAM bit plane is doubled in size to 256K bits and the video bit rate is doubled to 22 M bits/sec. The result is a much higher quality graphical representation minimizing the objectionable "Stair Cases" or "Jaggies".

Having high resolution is great, but you also must be able to update it quickly. The Tandy 2000 has a video graphics architecture using the new 16K x 4 RAM configuration that provides video memory updating between each 16 bit video word sent to the CRT instead of only during horizontal flyback as is common in other microcomputers. This results in a video memory read/write rate of over 22 M bits/sec which produces a full screen write in about 35 millisec.

Two more 256K video bit map memories are added in parallel to obtain in eight color 640 x 400 display. A palette RAM maps the 3 bits into any 4 bits so that any 8 of 16 colors can be displayed on the screen.

The character video has a 640 x 400 graphics mode and a special 800 x 400 text mode to achieve very high quality character shape and definition. The hardware provides all of the useful character attributes: reverse video, underline, blank, blink, intensify, and double wide-double high. If that is not enough, any 256 alternate character font can be loaded from disk into RAM to provide whatever style, type, or shape characters you desire, even foreign languages. Smooth scroll and split screen support in hardware are advanced features that are particularly useful in the new user friendly applications.

To display this high resolution graphics, Tandy has had a monochrome monitor with green phosphor and a custom designed color monitor that use the latest designs in electron guns and high contrast phosphors to achieve a bright, sharp, high fidelity display. Both of these monitors have a high speed horizontal scan rate of 26K Hz and video bandwidth of 25 MHz which is about 5 times more than color TV. What is really important is all of this high quality, high resolution, feature rich display capability has been achieved with very affordable costs - even with color. In a high performance workstation it is every bit as important to have a display that is pleasant and easy to view as it is to have applications that are easy to use.

The amount of on-line storage has always been a key parameter which limits a computer to solve a particular size application. The limit could be the number of people in your personnel file, the size of your spelling checker or the number of inventory records. The advent of integrated applications has made the capacity of the floppy disk drive even more critical, since all the applications as well as the data must be on-line. At least one integrated windowing environment will not function without using a hard disk. The floppy disk capacity is important even in a hard disk configuration in order to back-up on a reasonable number of floppy disks.

For this reason the Tandy 2000 uses the highest capacity floppy drives available in production quantities. These 96 track per inch, double-sided 5 1/4 floppy drives have a formatted capacity of 720K bytes per drive. That's 1,440,000 bytes of storage on-line with two drives. These half-height disk drives with direct drive brushless motors also have enhanced performance with twice the track-to-track step speed and all at only a small additional cost. On a total system cost per byte comparison, the IBM-PC costs almost two and one-half times more than the Tandy 2000.

The 10 megabyte built-in hard disk configuration with a new LSI hard disk controller is equally cost-effective.

We all know that without software a microcomputer is useless. The significant enhancements of three times the speed, twice the disk capacity, and twice the display resolution, makes it impossible to run IBM-PC application software that makes direct interface to the hardware even though the 8088 instruction set is executable. The amount of changes required in an applications program has been minimized by using MS-DOS with compatible system calls and bios entry points consistent with

the IBM-PC. The result is that hardware independent PC software that interfaces exclusively with the DOS and Bios will run directly on the Tandy 2000. However, many of the top tier PC programs interface directly to the hardware to compensate for 8088 performance and to provide copy protection and thus requires modification. Because of the high degree of compatibility, most software developers have found it to be a relative easy task to make the necessary changes to run on the Tandy 2000.

One major software supplier that required modification came to our labs, changed the code, and tested their software on the Tandy 2000 in about four hours. We are thus able to leverage the large amount of existing MS-DOS application software and simultaneously provide significant hardware enhancements.

I expect that you as reporters/security analysts have heard all the talk about technical details that you want to hear, but the engineering community will be continuing to talk a great deal about the Tandy TRS-80 Model 2000, the performance standard for the MS-DOS machines of the future.